Table 3.3 Proportions in clinical depression study for Exercise 11

	Treatment group (Y)			
Response (X)	Imipramine (1)	Lithium (2)	Combination (3)	Placebo (4)
Relapse (0)	0.120	0.087	0.146	0.160
No relapse (1)	0.147	0.166	0.107	0.067

indicator of the treatment group for that patient, and X, an indicator of whether or not the patient relapsed. Table 3.3 contains the joint p.f. of X and Y.

a. Calculate the probability that a patient selected at random from this study used Lithium (either alone or in combination with Imipramine) and did not relapse.

b. Calculate the probability that the patient had a relapse (without regard to the treatment group).

3.5 Marginal Distributions

Earlier in this chapter, we introduced distributions for random variables, and in Sec. 3.4 we discussed a generalization to joint distributions of two random variables simultaneously. Often, we start with a joint distribution of two random variables and we then want to find the distribution of just one of them. The distribution of one random variable X computed from a joint distribution is also called the marginal distribution of X. Each random variable will have a marginal c.d.f. as well as a marginal p.d.f. or p.f. We also introduce the concept of independent random variables, which is a natural generalization of independent events.

Deriving a Marginal p.f. or a Marginal p.d.f.

We have seen in Theorem 3.4.5 that if the joint c.d.f. F of two random variables X and Y is known, then the c.d.f. F_1 of the random variable X can be derived from F. We saw an example of this derivation in Example 3.4.15. If X has a continuous distribution, we can also derive the p.d.f. of X from the joint distribution.

Example 3.5.1

Demands for Utilities. Look carefully at the formula for $F(x, y)$ in Example 3.4.15, specifically the last two branches that we identified as $F_1(x)$ and $F_2(y)$, the c.d.f.'s of the two individual random variables X and Y. It is apparent from those two formulas and Theorem 3.3.5 that the p.d.f. of X alone is

$$f_1(x) = \begin{cases} \frac{1}{196} & \text{for } 4 \le x \le 200, \\ 0 & \text{otherwise,} \end{cases}$$

which matches what we already found in Example 3.2.1. Similarly, the p.d.f. of Y alone is

$$f_2(y) = \begin{cases} \frac{1}{149} & \text{for } 1 \le y \le 150, \\ 0 & \text{otherwise.} \end{cases}$$

◀

The ideas employed in Example 3.5.1 lead to the following definition.

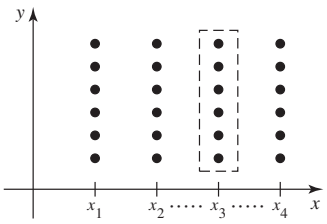


Figure 3.15 Computing $f_1(x)$ from the joint p.f.

Definition 3.5.1 Marginal c.d.f./p.f./p.d.f. Suppose that X and Y have a joint distribution. The c.d.f. of X derived by Theorem 3.4.5 is called the *marginal c.d.f.* of X. Similarly, the p.f. or p.d.f. of X associated with the marginal c.d.f. of X is called the *marginal p.f.* or *marginal p.d.f.* of X.

To obtain a specific formula for the marginal p.f. or marginal p.d.f., we start with a discrete joint distribution.

Theorem 3.5.1 If X and Y have a discrete joint distribution for which the joint p.f. is f, then the marginal p.f. f_1 of X is

$$f_1(x) = \sum_{\text{All } y} f(x, y). \tag{3.5.1}$$

Similarly, the marginal p.f. f_2 of Y is $f_2(y) = \sum_{\text{All } x} f(x, y)$.

Proof We prove the result for f_1, as the proof for f_2 is similar. We illustrate the proof in Fig. 3.15. In that figure, the set of points in the dashed box is the set of pairs with first coordinate x. The event $\{X = x\}$ can be expressed as the union of the events represented by the pairs in the dashed box, namely, $B_y = \{X = x \text{ and } Y = y\}$ for all possible y. The B_y events are disjoint and $\Pr(B_y) = f(x, y)$. Since $\Pr(X = x) = \sum_{\text{All } y} \Pr(B_y)$, Eq. (3.5.1) holds. ■

Example 3.5.2 Deriving a Marginal p.f. from a Table of Probabilities. Suppose that X and Y are the random variables in Example 3.4.3 on page 119. These are respectively the numbers of cars and televisions owned by a radomly selected household in a certain suburban area. Table 3.2 on page 119 gives their joint p.f., and we repeat that table in Table 3.4 together with row and column totals added to the margins.

The marginal p.f. f_1 of X can be read from the row totals of Table 3.4. The numbers were obtained by summing the values in each row of this table from the four columns in the central part of the table (those labeled $y = 1, 2, 3, 4$). In this way, it is found that $f_1(1) = 0.2$, $f_1(2) = 0.6$, $f_1(3) = 0.2$, and $f_1(x) = 0$ for all other values of x. This marginal p.f. gives the probabilities that a randomly selected household owns 1, 2, or 3 cars. Similarly, the marginal p.f. f_2 of Y, the probabilities that a household owns 1, 2, 3, or 4 televisions, can be read from the column totals. These numbers were obtained by adding the numbers in each of the columns from the three rows in the central part of the table (those labeled $x = 1, 2, 3$.) ◀

The name *marginal distribution* derives from the fact that the marginal distributions are the totals that appear in the margins of tables like Table 3.4.

If X and Y have a continuous joint distribution for which the joint p.d.f. is f, then the marginal p.d.f. f_1 of X is again determined in the manner shown in Eq. (3.5.1), but

Table 3.4 Joint p.f. $f(x, y)$ with marginal p.f.'s for Example 3.5.2

	y				
x	1	2	3	4	Total
1	0.1	0	0.1	0	0.2
2	0.3	0	0.1	0.2	0.6
3	0	0.2	0	0	0.2
Total	0.4	0.2	0.2	0.2	1.0

the sum over all possible values of Y is now replaced by the integral over all possible values of Y.

Theorem 3.5.2 If X and Y have a continuous joint distribution with joint p.d.f. f, then the marginal p.d.f. f_1 of X is

$$f_1(x) = \int_{-\infty}^{\infty} f(x, y)\, dy \quad \text{for } -\infty < x < \infty. \tag{3.5.2}$$

Similarly, the marginal p.d.f. f_2 of Y is

$$f_2(y) = \int_{-\infty}^{\infty} f(x, y)\, dx \quad \text{for } -\infty < y < \infty. \tag{3.5.3}$$

Proof We prove (3.5.2) as the proof of (3.5.3) is similar. For each x, $\Pr(X \le x)$ can be written as $\Pr((X, Y) \in C)$, where $C = \{(r, s)) : r \le x\}$. We can compute this probability directly from the joint p.d.f. of X and Y as

$$\begin{aligned} \Pr((X, Y) \in C) &= \int_{-\infty}^{x} \int_{-\infty}^{\infty} f(r, s) ds dr \\ &= \int_{-\infty}^{x} \left[\int_{-\infty}^{\infty} f(r, s) ds \right] dr \end{aligned} \tag{3.5.4}$$

The inner integral in the last expression of Eq. (3.5.4) is a function of r and it can easily be recognized as $f_1(r)$, where f_1 is defined in Eq. (3.5.2). It follows that $\Pr(X \le x) = \int_{-\infty}^{x} f_1(r) dr$, so f_1 is the marginal p.d.f. of X. ■

Example 3.5.3 Deriving a Marginal p.d.f. Suppose that the joint p.d.f. of X and Y is as specified in Example 3.4.8, namely,

$$f(x, y) = \begin{cases} \frac{21}{4} x^2 y & \text{for } x^2 \le y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

The set S of points (x, y) for which $f(x, y) > 0$ is sketched in Fig. 3.16. We shall determine first the marginal p.d.f. f_1 of X and then the marginal p.d.f. f_2 of Y.

It can be seen from Fig. 3.16 that X cannot take any value outside the interval $[-1, 1]$. Therefore, $f_1(x) = 0$ for $x < -1$ or $x > 1$. Furthermore, for $-1 \le x \le 1$, it is seen from Fig. 3.16 that $f(x, y) = 0$ unless $x^2 \le y \le 1$. Therefore, for $-1 \le x \le 1$,

$$f_1(x) = \int_{-\infty}^{\infty} f(x, y)\, dy = \int_{x^2}^{1} \left(\frac{21}{4} \right) x^2 y\, dy = \left(\frac{21}{8} \right) x^2 (1 - x^4).$$

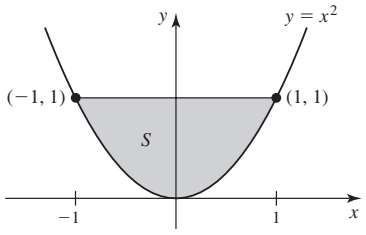


Figure 3.16 The set S where $f(x, y) > 0$ in Example 3.5.3.

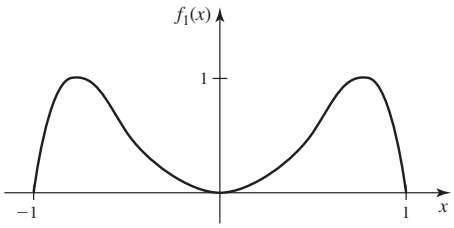


Figure 3.17 The marginal p.d.f. of X in Example 3.5.3.

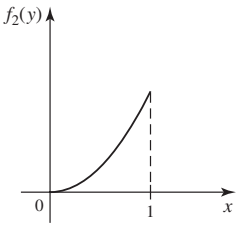


Figure 3.18 The marginal p.d.f. of Y in Example 3.5.3.

This marginal p.d.f. of X is sketched in Fig. 3.17.

Next, it can be seen from Fig. 3.16 that Y cannot take any value outside the interval $[0, 1]$. Therefore, $f_2(y) = 0$ for $y < 0$ or $y > 1$. Furthermore, for $0 \le y \le 1$, it is seen from Fig. 3.12 that $f(x, y) = 0$ unless $-\sqrt{y} \le x \le \sqrt{y}$. Therefore, for $0 \le y \le 1$,

$$f_2(y) = \int_{-\infty}^{\infty} f(x, y)\, dx = \int_{-\sqrt{y}}^{\sqrt{y}} \left(\frac{21}{4}\right) x^2 y\, dx = \left(\frac{7}{2}\right) y^{5/2}.$$

This marginal p.d.f. of Y is sketched in Fig. 3.18. ◀

If X has a discrete distribution and Y has a continuous distribution, we can derive the marginal p.f. of X and the marginal p.d.f. of Y from the joint p.f./p.d.f. in the same ways that we derived a marginal p.f. or a marginal p.d.f. from a joint p.f. or a joint p.d.f. The following result can be proven by combining the techniques used in the proofs of Theorems 3.5.1 and 3.5.2.

Theorem 3.5.3

Let f be the joint p.f./p.d.f. of X and Y, with X discrete and Y continuous. Then the marginal p.f. of X is

$$f_1(x) = \Pr(X = x) = \int_{-\infty}^{\infty} f(x, y)\, dy, \quad \text{for all } x,$$

and the marginal p.d.f. of Y is

$$f_2(y) = \sum_x f(x, y), \quad \text{for } -\infty < y < \infty. \qquad \blacksquare$$

Example 3.5.4 Determining a Marginal p.f. and Marginal p.d.f. from a Joint p.f./p.d.f. Suppose that the joint p.f./p.d.f. of X and Y is as in Example 3.4.11 on page 124. The marginal p.f. of X is obtained by integrating

$$f_1(x) = \int_0^1 \frac{xy^{x-1}}{3} dy = \frac{1}{3},$$

for $x = 1, 2, 3$. The marginal p.d.f. of Y is obtained by summing

$$f_2(y) = \frac{1}{3} + \frac{2y}{3} + y^2, \quad \text{for } 0 < y < 1. \qquad \blacktriangleleft$$

Although the marginal distributions of X and Y can be derived from their joint distribution, it is not possible to reconstruct the joint distribution of X and Y from their marginal distributions without additional information. For instance, the marginal p.d.f.'s sketched in Figs. 3.17 and 3.18 reveal no information about the relationship between X and Y. In fact, by definition, the marginal distribution of X specifies probabilities for X without regard for the values of any other random variables. This property of a marginal p.d.f. can be further illustrated by another example.

Example 3.5.5 Marginal and Joint Distributions. Suppose that a penny and a nickel are each tossed n times so that every pair of sequences of tosses (n tosses in each sequence) is equally likely to occur. Consider the following two definitions of X and Y: (i) X is the number of heads obtained with the penny, and Y is the number of heads obtained with the nickel. (ii) Both X and Y are the number of heads obtained with the penny, so the random variables X and Y are actually identical.

In case (i), the marginal distribution of X and the marginal distribution of Y will be identical binomial distributions. The same pair of marginal distributions of X and Y will also be obtained in case (ii). However, the joint distribution of X and Y will not be the same in the two cases. In case (i), X and Y can take different values. Their joint p.f. is

$$f(x, y) = \begin{cases} \binom{n}{x}\binom{n}{y}\left(\frac{1}{2}\right)^{x+y} & \text{for } x = 0, 1 \ldots, n,\ y = 0, 1, \ldots, n, \\ 0 & \text{otherwise.} \end{cases}$$

In case (ii), X and Y must take the same value, and their joint p.f. is

$$f(x, y) = \begin{cases} \binom{n}{x}\left(\frac{1}{2}\right)^{x} & \text{for } x = y = 0, 1 \ldots, n, \\ 0 & \text{otherwise.} \end{cases} \qquad \blacktriangleleft$$

Independent Random Variables

Example 3.5.6 Demands for Utilities. In Examples 3.4.15 and 3.5.1, we found the marginal c.d.f.'s of water and electric demand were, respectively,

$$F_1(x) = \begin{cases} 0 & \text{for } x < 4, \\ \frac{x}{196} & \text{for } 4 \le x \le 200, \\ 1 & \text{for } x > 200, \end{cases} \qquad F_2(y) = \begin{cases} 0 & \text{for } y < 1, \\ \frac{y}{149} & \text{for } 1 \le y \le 150, \\ 1 & \text{for } y > 150. \end{cases}$$

The product of these two functions is precisely the same as the joint c.d.f. of X and Y given in Example 3.5.1. One consequence of this fact is that, for every x and y, $\Pr(X \le x, \text{ and } Y \le y) = \Pr(X \le x)\Pr(Y \le y)$. This equation makes X and Y an example of the next definition. ◀

Definition 3.5.2 Independent Random Variables. It is said that two random variables X and Y are *independent* if, for every two sets A and B of real numbers such that $\{X \in A\}$ and $\{Y \in B\}$ are events,

$$\Pr(X \in A \text{ and } Y \in B) = \Pr(X \in A)\Pr(Y \in B). \tag{3.5.5}$$

In other words, let E be any event the occurrence or nonoccurrence of which depends only on the value of X (such as $E = \{X \in A\}$), and let D be any event the occurrence or nonoccurrence of which depends only on the value of Y (such as $D = \{Y \in B\}$). Then X and Y are independent random variables if and only if E and D are independent events for all such events E and D.

If X and Y are independent, then for all real numbers x and y, it must be true that

$$\Pr(X \le x \text{ and } Y \le y) = \Pr(X \le x)\Pr(Y \le y). \tag{3.5.6}$$

Moreover, since all probabilities for X and Y of the type appearing in Eq. (3.5.5) can be derived from probabilities of the type appearing in Eq. (3.5.6), it can be shown that if Eq. (3.5.6) is satisfied for all values of x and y, then X and Y must be independent. The proof of this statement is beyond the scope of this book and is omitted, but we summarize it as the following theorem.

Theorem 3.5.4 Let the joint c.d.f. of X and Y be F, let the marginal c.d.f. of X be F_1, and let the marginal c.d.f. of Y be F_2. Then X and Y are independent if and only if, for all real numbers x and y, $F(x, y) = F_1(x)F_2(y)$. ■

For example, the demands for water and electricity in Example 3.5.6 are independent. If one returns to Example 3.5.1, one also sees that the product of the marginal p.d.f.'s of water and electric demand equals their joint p.d.f. given in Eq. (3.4.2). This relation is characteristic of independent random variables whether discrete or continuous.

Theorem 3.5.5 Suppose that X and Y are random variables that have a joint p.f., p.d.f., or p.f./p.d.f. f. Then X and Y will be independent if and only if f can be represented in the following form for $-\infty < x < \infty$ and $-\infty < y < \infty$:

$$f(x, y) = h_1(x)h_2(y), \tag{3.5.7}$$

where h_1 is a nonnegative function of x alone and h_2 is a nonnegative function of y alone.

Proof We shall give the proof only for the case in which X is discrete and Y is continuous. The other cases are similar. For the "if" part, assume that Eq. (3.5.7) holds. Write

$$f_1(x) = \int_{-\infty}^{\infty} h_1(x)h_2(y)dy = c_1h_1(x),$$

where $c_1 = \int_{-\infty}^{\infty} h_2(y)dy$ must be finite and strictly positive, otherwise f_1 wouldn't be a p.f. So, $h_1(x) = f_1(x)/c_1$. Similarly,

$$f_2(y) = \sum_x h_1(x)h_2(y) = h_2(y)\sum_x \frac{1}{c_1}f_1(x) = \frac{1}{c_1}h_2(y).$$

So, $h_2(y) = c_1 f_2(y)$. Since $f(x, y) = h_1(x)h_2(y)$, it follows that

$$f(x, y) = \frac{f_1(x)}{c_1} c_1 f_2(y) = f_1(x) f_2(y). \tag{3.5.8}$$

Now let A and B be sets of real numbers. Assuming the integrals exist, we can write

$$\begin{aligned}\Pr(X \in A \text{ and } Y \in B) &= \sum_{x \in A} \int_B f(x, y) dy \\ &= \int_B \sum_{x \in A} f_1(x) f_2(y) dy, \\ &= \sum_{x \in A} f_1(x) \int_B f_2(y) dy,\end{aligned}$$

where the first equality is from Definition 3.4.5, the second is from Eq. (3.5.8), and the third is straightforward rearrangement. We now see that X and Y are independent according to Definition 3.5.2.

For the "only if" part, assume that X and Y are independent. Let A and B be sets of real numbers. Let f_1 be the marginal p.d.f. of X, and let f_2 be the marginal p.f. of Y. Then

$$\begin{aligned}\Pr(X \in A \text{ and } Y \in B) &= \sum_{x \in A} f_1(x) \int_B f_2(y) dy \\ &= \int_B \sum_{x \in A} f_1(x) f_2(y) dy,\end{aligned}$$

(if the integral exists) where the first equality follows from Definition 3.5.2 and the second is a straightforward rearrangement. We now see that $f_1(x) f_2(y)$ satisfies the conditions needed to be $f(x, y)$ as stated in Definition 3.4.5. ■

A simple corollary follows from Theorem 3.5.5.

Corollary 3.5.1 Two random variables X and Y are independent if and only if the following factorization is satisfied for all real numbers x and y:

$$f(x, y) = f_1(x) f_2(y). \tag{3.5.9}$$

■

As stated in Sec. 3.2 (see page 102), in a continuous distribution the values of a p.d.f. can be changed arbitrarily at any countable set of points. Therefore, for such a distribution it would be more precise to state that the random variables X and Y are independent if and only if it is possible to choose versions of f, f_1, and f_2 such that Eq. (3.5.9) is satisfied for $-\infty < x < \infty$ and $-\infty < y < \infty$.

The Meaning of Independence We have given a mathematical definition of independent random variables in Definition 3.5.2, but we have not yet given any interpretation of the concept of independent random variables. Because of the close connection between independent events and independent random variables, the interpretation of independent random variables should be closely related to the interpretation of independent events. We model two events as independent if learning that one of them occurs does not change the probability that the other one occurs. It is easiest to extend this idea to discrete random variables. Suppose that X and Y

Table 3.5 Joint p.f. $f(x, y)$ for Example 3.5.7

	y						
x	1	2	3	4	5	6	Total
0	1/24	1/24	1/24	1/24	1/24	1/24	1/4
1	1/12	1/12	1/12	1/12	1/12	1/12	1/2
2	1/24	1/24	1/24	1/24	1/24	1/24	1/4
Total	1/6	1/6	1/6	1/6	1/6	1/6	1.000

have a discrete joint distribution. If, for each y, learning that $Y = y$ does not change any of the probabilities of the events $\{X = x\}$, we would like to say that X and Y are independent. From Corollary 3.5.1 and the definition of marginal p.f., we see that indeed X and Y are independent if and only if, for each y and x such that $\Pr(Y = y) > 0$, $\Pr(X = x|Y = y) = \Pr(X = x)$, that is, learning the value of Y doesn't change any of the probabilities associated with X. When we formally define conditional distributions in Sec. 3.6, we shall see that this interpretation of independent discrete random variables extends to all bivariate distributions. In summary, if we are trying to decide whether or not to model two random variables X and Y as independent, we should think about whether we would change the distribution of X after we learned the value of Y or vice versa.

Example 3.5.7

Games of Chance. A carnival game consists of rolling a fair die, tossing a fair coin two times, and recording both outcomes. Let Y stand for the number on the die, and let X stand for the number of heads in the two tosses. It seems reasonable to believe that all of the events determined by the roll of the die are independent of all of the events determined by the flips of the coin. Hence, we can assume that X and Y are independent random variables. The marginal distribution of Y is the uniform distribution on the integers $1, \ldots, 6$, while the distribution of X is the binomial distribution with parameters 2 and 1/2. The marginal p.f.'s and the joint p.f. of X and Y are given in Table 3.5, where the joint p.f. was constructed using Eq. (3.5.9). The Total column gives the marginal p.f. f_1 of X, and the Total row gives the marginal p.f. f_2 of Y. ◀

Example 3.5.8

Determining Whether Random Variables Are Independent in a Clinical Trial. Return to the clinical trial of depression drugs in Exercise 11 of Sec. 3.4 (on page 129). In that trial, a patient is selected at random from the 150 patients in the study and we record Y, an indicator of the treatment group for that patient, and X, an indicator of whether or not the patient relapsed. Table 3.6 repeats the joint p.f. of X and Y along with the marginal distributions in the margins. We shall determine whether or not X and Y are independent.

In Eq. (3.5.9), $f(x, y)$ is the probability in the xth row and the yth column of the table, $f_1(x)$ is the number in the Total column in the xth row, and $f_2(y)$ is the number in the Total row in the yth column. It is seen in the table that $f(1, 2) = 0.087$, while $f_1(1) = 0.513$, and $f_2(1) = 0.253$. Hence, $f(1, 2) \neq f_1(1)f_2(1) = 0.129$. It follows that X and Y are not independent. ◀

It should be noted from Examples 3.5.7 and 3.5.8 that X and Y will be independent if and only if the rows of the table specifying their joint p.f. are proportional to

Table 3.6 Proportions marginals in Example 3.5.8

	Treatment group (Y)				
Response (X)	Imipramine (1)	Lithium (2)	Combination (3)	Placebo (4)	Total
Relapse (0)	0.120	0.087	0.146	0.160	0.513
No relapse (1)	0.147	0.166	0.107	0.067	0.487
Total	0.267	0.253	0.253	0.227	1.0

one another, or equivalently, if and only if the columns of the table are proportional to one another.

Example 3.5.9 Calculating a Probability Involving Independent Random Variables. Suppose that two measurements X and Y are made of the rainfall at a certain location on May 1 in two consecutive years. It might be reasonable, given knowledge of the history of rainfall on May 1, to treat the random variables X and Y as independent. Suppose that the p.d.f. g of each measurement is as follows:

$$g(x) = \begin{cases} 2x & \text{for } 0 \le x \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

We shall determine the value of $\Pr(X + Y \le 1)$.

Since X and Y are independent and each has the p.d.f. g, it follows from Eq. (3.5.9) that for all values of x and y the joint p.d.f. $f(x, y)$ of X and Y will be specified by the relation $f(x, y) = g(x)g(y)$. Hence,

$$f(x, y) = \begin{cases} 4xy & \text{for } 0 \le x \le 1 \text{ and } 0 \le y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

The set S in the xy-plane, where $f(x, y) > 0$, and the subset S_0, where $x + y \le 1$, are sketched in Fig. 3.19. Thus,

$$\Pr(X + Y \le 1) = \int_{S_0} \int f(x, y)\, dx\, dy = \int_0^1 \int_0^{1-x} 4xy\, dy\, dx = \frac{1}{6}.$$

As a final note, if the two measurements X and Y had been made on the same day at nearby locations, then it might not make as much sense to treat them as independent, since we would expect them to be more similar to each other than to historical rainfalls. For example, if we first learn that X is small compared to historical rainfall on the date in question, we might then expect Y to be smaller than the historical distribution would suggest. ◀

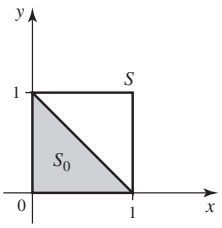


Figure 3.19 The subset S_0 where $x + y \le 1$ in Example 3.5.9.

Theorem 3.5.5 says that X and Y are independent if and only if, for all values of x and y, f can be factored into the product of an arbitrary nonnegative function of x and an arbitrary nonnegative function of y. However, it should be emphasized that, just as in Eq. (3.5.9), the factorization in Eq. (3.5.7) must be satisfied for all values of x and y ($-\infty < x < \infty$ and $-\infty < y < \infty$).

Example 3.5.10 Dependent Random Variables. Suppose that the joint p.d.f. of X and Y has the following form:

$$f(x, y) = \begin{cases} kx^2y^2 & \text{for } x^2 + y^2 \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

We shall show that X and Y are not independent.

It is evident that at each point inside the circle $x^2 + y^2 \le 1$, $f(x, y)$ can be factored as in Eq. (3.5.7). However, this same factorization cannot also be satisfied at every point outside this circle. For example, $f(0.9, 0.9) = 0$, but neither $f_1(0.9) = 0$ nor $f_2(0.9) = 0$. (In Exercise 13, you can verify this feature of f_1 and f_2.)

The important feature of this example is that the values of X and Y are constrained to lie inside a circle. The joint p.d.f. of X and Y is positive inside the circle and zero outside the circle. Under these conditions, X and Y cannot be independent, because for every given value y of Y, the possible values of X will depend on y. For example, if $Y = 0$, then X can have any value such that $X^2 \le 1$; if $Y = 1/2$, then X must have a value such that $X^2 \le 3/4$. ◀

Example 3.5.10 shows that one must be careful when trying to apply Theorem 3.5.5. The situation that arose in that example will occur whenever $\{(x, y) : f(x, y) > 0\}$ has boundaries that are curved or not parallel to the coordinate axes. There is one important special case in which it is easy to check the conditions of Theorem 3.5.5. The proof is left as an exercise.

Theorem 3.5.6 Let X and Y have a continuous joint distribution. Suppose that $\{(x, y) : f(x, y) > 0\}$ is a rectangular region R (possibly unbounded) with sides (if any) parallel to the coordinate axes. Then X and Y are independent if and only if Eq. (3.5.7) holds for all $(x, y) \in R$. ■

Example 3.5.11 Verifying the Factorization of a Joint p.d.f. Suppose that the joint p.d.f. f of X and Y is as follows:

$$f(x, y) = \begin{cases} ke^{-(x+2y)} & \text{for } x \ge 0 \text{ and } y \ge 0, \\ 0 & \text{otherwise,} \end{cases}$$

where k is some constant. We shall first determine whether X and Y are independent and then determine their marginal p.d.f.'s.

In this example, $f(x, y) = 0$ outside of an unbounded rectangular region R whose sides are the lines $x = 0$ and $y = 0$. Furthermore, at each point inside R, $f(x, y)$ can be factored as in Eq. (3.5.7) by letting $h_1(x) = ke^{-x}$ and $h_2(y) = e^{-2y}$. Therefore, X and Y are independent.

It follows that in this case, except for constant factors, $h_1(x)$ for $x \ge 0$ and $h_2(y)$ for $y \ge 0$ must be the marginal p.d.f.'s of X and Y. By choosing constants that make $h_1(x)$ and $h_2(y)$ integrate to unity, we can conclude that the marginal p.d.f.'s f_1 and f_2 of X and Y must be as follows:

$$f_1(x) = \begin{cases} e^{-x} & \text{for } x \ge 0, \\ 0 & \text{otherwise,} \end{cases}$$

and

$$f_2(y) = \begin{cases} 2e^{-2y} & \text{for } y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

If we multiply $f_1(x)$ times $f_2(y)$ and compare the product to $f(x, y)$, we see that $k = 2$. ◀

Note: Separate Functions of Independent Random Variables Are Independent. If X and Y are independent, then $h(X)$ and $g(Y)$ are independent no matter what the functions h and g are. This is true because for every t, the event $\{h(X) \leq t\}$ can always be written as $\{X \in A\}$, where $A = \{x : h(x) \leq t\}$. Similarly, $\{g(Y) \leq u\}$ can be written as $\{Y \in B\}$, so Eq. (3.5.6) for $h(X)$ and $g(Y)$ follows from Eq. (3.5.5) for X and Y.

Summary

Let $f(x, y)$ be a joint p.f., joint p.d.f., or joint p.f./p.d.f. of two random variables X and Y. The marginal p.f. or p.d.f. of X is denoted by $f_1(x)$, and the marginal p.f. or p.d.f. of Y is denoted by $f_2(y)$. To obtain $f_1(x)$, compute $\sum_y f(x, y)$ if Y is discrete or $\int_{-\infty}^{\infty} f(x, y)\, dy$ if Y is continuous. Similarly, to obtain $f_2(y)$, compute $\sum_x f(x, y)$ if X is discrete or $\int_{-\infty}^{\infty} f(x, y)\, dx$ if X is continuous. The random variables X and Y are independent if and only if $f(x, y) = f_1(x) f_2(y)$ for *all* x and y. This is true regardless of whether X and/or Y is continuous or discrete. A sufficient condition for two continuous random variables to be independent is that $R = \{(x, y) : f(x, y) > 0\}$ be rectangular with sides parallel to the coordinate axes and that $f(x, y)$ factors into separate functions of x of y in R.

Exercises

1. Suppose that X and Y have a continuous joint distribution for which the joint p.d.f. is

$$f(x, y) = \begin{cases} k & \text{for } a \leq x \leq b \text{ and } c \leq y \leq d, \\ 0 & \text{otherwise,} \end{cases}$$

where $a < b$, $c < d$, and $k > 0$. Find the marginal distributions of X and Y.

2. Suppose that X and Y have a discrete joint distribution for which the joint p.f. is defined as follows:

$$f(x, y) = \begin{cases} \frac{1}{30}(x + y) & \text{for } x = 0, 1, 2 \text{ and } y = 0, 1, 2, 3, \\ 0 & \text{otherwise.} \end{cases}$$

a. Determine the marginal p.f.'s of X and Y.

b. Are X and Y independent?

3. Suppose that X and Y have a continuous joint distribution for which the joint p.d.f. is defined as follows:

$$f(x, y) = \begin{cases} \frac{3}{2}y^2 & \text{for } 0 \leq x \leq 2 \text{ and } 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

a. Determine the marginal p.d.f.'s of X and Y.

b. Are X and Y independent?

c. Are the event $\{X < 1\}$ and the event $\{Y \geq 1/2\}$ independent?

4. Suppose that the joint p.d.f. of X and Y is as follows:

$$f(x, y) = \begin{cases} \frac{15}{4}x^2 & \text{for } 0 \leq y \leq 1 - x^2, \\ 0 & \text{otherwise.} \end{cases}$$

a. Determine the marginal p.d.f.'s of X and Y.

b. Are X and Y independent?

5. A certain drugstore has three public telephone booths. For $i = 0, 1, 2, 3$, let p_i denote the probability that exactly i telephone booths will be occupied on any Monday evening at 8:00 P.M.; and suppose that $p_0 = 0.1$, $p_1 = 0.2$, $p_2 = 0.4$, and $p_3 = 0.3$. Let X and Y denote the number of booths that will be occupied at 8:00 P.M. on two independent Monday evenings. Determine: **(a)** the joint p.f. of X and Y; **(b)** $\Pr(X = Y)$; **(c)** $\Pr(X > Y)$.

6. Suppose that in a certain drug the concentration of a particular chemical is a random variable with a continuous distribution for which the p.d.f. g is as follows:

$$g(x) = \begin{cases} \frac{3}{8}x^2 & \text{for } 0 \le x \le 2, \\ 0 & \text{otherwise.} \end{cases}$$

Suppose that the concentrations X and Y of the chemical in two separate batches of the drug are independent random variables for each of which the p.d.f. is g. Determine **(a)** the joint p.d.f. of X and Y; **(b)** $\Pr(X = Y)$; **(c)** $\Pr(X > Y)$; **(d)** $\Pr(X + Y \le 1)$.

7. Suppose that the joint p.d.f. of X and Y is as follows:

$$f(x, y) = \begin{cases} 2xe^{-y} & \text{for } 0 \le x \le 1 \text{ and } 0 < y < \infty, \\ 0 & \text{otherwise.} \end{cases}$$

Are X and Y independent?

8. Suppose that the joint p.d.f. of X and Y is as follows:

$$f(x, y) = \begin{cases} 24xy & \text{for } x \ge 0,\ y \ge 0, \text{ and } x + y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Are X and Y independent?

9. Suppose that a point (X, Y) is chosen at random from the rectangle S defined as follows:

$$S = \{(x, y) : 0 \le x \le 2 \text{ and } 1 \le y \le 4\}.$$

a. Determine the joint p.d.f. of X and Y, the marginal p.d.f. of X, and the marginal p.d.f. of Y.

b. Are X and Y independent?

10. Suppose that a point (X, Y) is chosen at random from the circle S defined as follows:

$$S = \{(x, y) : x^2 + y^2 \le 1\}.$$

a. Determine the joint p.d.f. of X and Y, the marginal p.d.f. of X, and the marginal p.d.f. of Y.

b. Are X and Y independent?

11. Suppose that two persons make an appointment to meet between 5 P.M. and 6 P.M. at a certain location, and they agree that neither person will wait more than 10 minutes for the other person. If they arrive independently at random times between 5 P.M. and 6 P.M., what is the probability that they will meet?

12. Prove Theorem 3.5.6.

13. In Example 3.5.10, verify that X and Y have the same marginal p.d.f.'s and that

$$f_1(x) = \begin{cases} 2kx^2(1 - x^2)^{3/2}/3 & \text{if } -1 \le x \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

14. For the joint p.d.f. in Example 3.4.7, determine whether or not X and Y are independent.

15. A painting process consists of two stages. In the first stage, the paint is applied, and in the second stage, a protective coat is added. Let X be the time spent on the first stage, and let Y be the time spent on the second stage. The first stage involves an inspection. If the paint fails the inspection, one must wait three minutes and apply the paint again. After a second application, there is no further inspection. The joint p.d.f. of X and Y is

$$f(x, y) = \begin{cases} \frac{1}{3} & \text{if } 1 < x < 3 \text{ and } 0 < y < 1, \\ \frac{1}{6} & \text{if } 6 < x < 8 \text{ and } 0 < y < 1, \\ 0 & \text{otherwise.} \end{cases}$$

a. Sketch the region where $f(x, y) > 0$. Note that it is not exactly a rectangle.

b. Find the marginal p.d.f.'s of X and Y.

c. Show that X and Y are independent.

This problem does not contradict Theorem 3.5.6. In that theorem the conditions, including that the set where $f(x, y) > 0$ be rectangular, are sufficient but not necessary.

3.6 Conditional Distributions

We generalize the concept of conditional probability to conditional distributions. Recall that distributions are just collections of probabilities of events determined by random variables. Conditional distributions will be the probabilities of events determined by some random variables conditional on events determined by other random variables. The idea is that there will typically be many random variables of interest in an applied problem. After we observe some of those random variables, we want to be able to adjust the probabilities associated with the ones that have not yet been observed. The conditional distribution of one random variable X given another Y will be the distribution that we would use for X after we learn the value of Y.